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ForeFire: A Modular, Scriptable C++ Simulation Engine and Library for Wildland-Fire Spread

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Software

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Summary

Wildfire forecasting is at the same time an active research area and an important need for operational decision support systems. **ForeFire** is a modular, high-performance wildland fire simulation engine implemented in C++. It is designed to model the spread of wildfire perimeters over large landscapes at meter scale resolution, serving both as a research platform and an operational forecasting tool. It also offers multiple interfaces, the core C++ library has Fortran and Python bindings and is accompanied by a lightweight scriptable interpreter (in custom FF language), a local HTTP service that enables a customizable graphical user interface and ability to load, save and export data in NetCDF, GeoJson, KML, png and jpg. Simulation is discrete-event-driven (Filippi et al., 2009), focusing computational effort on the active region of the fire front defined as a dynamic mesh or multipolygons of fire markers. ForeFire is also capable of two-way coupling with the MesoNH (Lac et al., 2018) atmospheric model to account for fire-atmosphere interaction (Filippi et al., 2013).

Statement of need

Wildfire modeling tools have historically been split between **complex combustion research models** and **streamlined operational tools**, each with distinct limitations. On one end of the spectrum, computational combustion and fluid dynamics (CFD) based models (e.g., FIRETEC (Linn & Cunningham, 2005) or WFDS (Mell et al., 2007)) provide detailed physics with complete control of combustion models at flame scale but are highly computationally intensive and yet unable to provide faster than real time large wildfire forecasting. Another family of wildland fire spread solvers such as WRF/SFire (Mandel et al., 2011) can resolve coupled front propagation and local meteorology but must be run within an atmospheric model that requires a large amount of processing power and data.

On the other end, operational wildfire simulators built for practitioners, such as widely used Farsite (now flammap) (Finney, 1998) or Canadian Prometheus (Garcia et al., 2008) are able to simulate tens of kilometers long fire front in matter of seconds, prioritizing speed and simplicity at the expense of generality. These codes have definite built-in modeling assumptions and are distributed (freely) as compiled software with graphical interfaces and limited access to source code. Existing open-source libraries, such as ElmFire (Lautenberger, 2013) or Cell2Fire (Pais et al., 2021), are each built around a single core spread model and are packaged with streamlined, simplified parameterization to prioritize practitioner efficiency. While being well-packaged is advantageous for direct use, it makes it difficult for researchers to implement new modeling approaches or to streamline and script integration into different experimental frameworks. This gap between highly complex customizable models and more rigid operational tools creates a

need for a **unified solution** : a wildfire simulator that is both **adaptable** and **high-performing**. ForeFire was developed to fill this need by bridging the flexibility of research-oriented code with the efficiency and user-friendliness of operational systems (Filippi et al., 2018). ForeFire's modular architecture and scripting interface allow users to easily swap or implement new fire models without changing the core code and test multiple simulation scenarions with same input files. It can also run fast in a standalone mode (for purely surface fire simulations) or be coupled with atmospheric models, all under a consistent interface and parameterization. ForeFire's support for **multiple language bindings** (HTTP, C++ API, Python/NumPy, Fortran) is also unique and further expands its usability across different domains. These capabilities enable rapid prototyping and experimentation on large datesets, and also support operational needs by allowing integration into decision support pipelines including coupled with atmospheric models.

Some typical Use Cases

Rapid prototyping of new spread or flux models

The software also implements several standard fire fluxes and spread rate models, including different variants of the widely used semi-empirical Rothermel model (Andrews, 2018) as well as Balbi semi-physical formulation (Balbi et al., 2009), with an architecture that makes it trivial to switch, extend or add to this model base by adding a single `.cpp` using any existing model file as template. Internally data is handled as *layers* that can come from Python NumPy array, supplied from input NetCDF files or generated on the fly by ForeFire (e.g. slope derived from the elevation layer). Developping a Rate Of Spread wildfire model was the original purpose of this simulation code and helped to iterate versions the Balbi Rate Of Spread formulation on real-cases studies in (Balbi et al., 2009) and (Santoni et al., 2011). It also served to implement various heat and chemical species flux models used for volcanic eruption in (Filippi et al., 2021), wildfire plume chemical compounds in (Strada et al., 2012) or industrial fires in (Baggio et al., 2022). In addition, the code includes a generic `ANNPropagationModel`, which implements a feedforward artificial neural network (ANN). This model expects a pre-trained network in the form of a graph input file to replace any ROS or flux functions.

Batch simulations with the ForeFire scripting

Custom FF language allows to easily generate multiple scenarios, including fire-fighting strategies or ensemble forecasts, with light scripting efforts. The interpreter run these scripts either from a file or from another process (piped trough the shell). A FF script is a set of instructions that are interpreted at a specific date, updated with a `stepor` a `goTo` commands or scheduled if post-fixed by a `@t=` operator.

Each of these commands (such as, `goTo[t=42]`, `include[state.ff]`, `startFire[lonlat=(-8.1, 39.9,0)]@t=42`, `setParameter[propagationModel=Rothermel]` or `plot[parameter=speed;filena` of ForeFire is bound or can directly be called from HTTP, C++, Fortran or Python, and are the core logic of the library. Help messages and autocompletion are directly available from the shell interpreter that can be run interactively, with several examples provided in the repository. ForeFire can also be run interactively using a highly customizable web-based graphical interface (command `listenHTTP[host:port]`), that starts a local HTTP service with ForeFire bindings and serves standard or user-defined web pages as shown in Figure 1.

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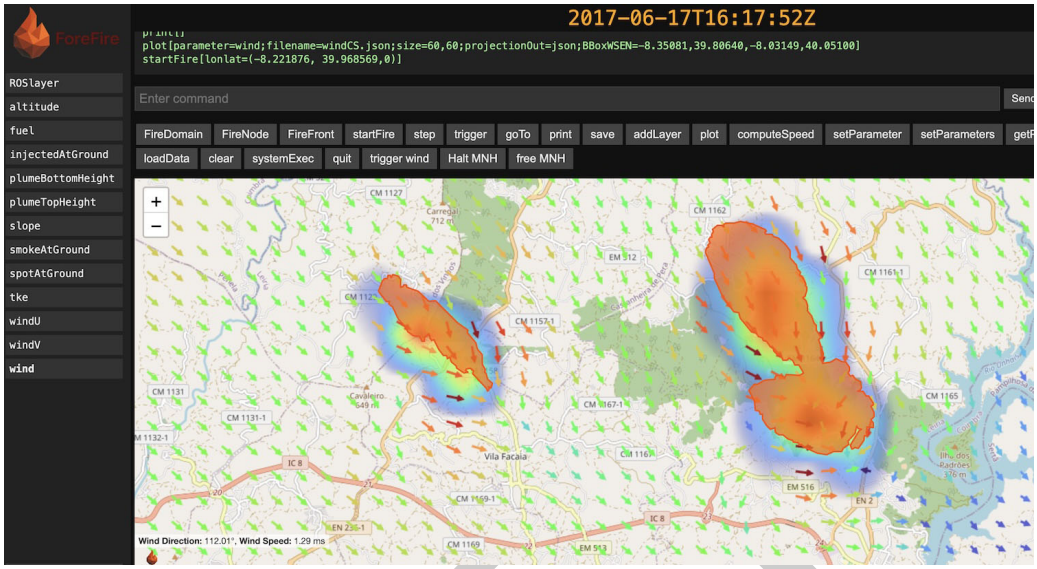

Figure 1: Default web interface accessed trough ForeFire internal HTTP service and exposing interactive commands. The simulation here is a wildfire in Portugal with ForeFire coupled with an atmospheric model that provides winds.

By utilizing pre-compiled datasets over extensive regions, this approach supports continent-wide operational forecasting services. It has been applied to identify optimal escape routes (Kamilaris et al., 2023), integrated into the French national WildFire Decision Support System OPEN DFCI, showcased on the FireCaster demonstration platform, and also currently used in commercial simulation services AriaFire Firecaster, UmGraueMeio Pantera and Ororatech FireSpread.

The ability to run as batch also means it can be used to perform systematic model evaluation (Filippi et al., 2014) generate simulation ensembles (Allaire et al., 2020) or large simulation databases for deep learning (Allaire et al., 2021),(Allaire et al., 2022).

Two-way coupling with the MesoNH atmospheric model

Once tested as standalone simulations, same scripts and input data can be executed in a coupled mode with the Open-Source atmospheric model MesoNH (Lac et al., 2018). In this configuration MesoNH executes ForeFire (FORTRAN Bindings) as an MPI sub-process with fire propagation using surface fields (wind) from the atmospheric model and distributes fluxes fields to the atmosphic model. Coupled simulation can be run on a laptop at field scale to evaluate model ability on prescribed well instrumented experiments (Filippi et al., 2013), but also on hundreds of processors to perform faster than real-time high resolution simulation of fire-induced winds on a large wildfire (Filippi et al., 2018), fire-induced convection during an extreme wildfire event (Couto et al., 2024),(Campos et al., 2023) or even to estimate wildfire spotting (firebrands) with a resolved fire plume (Alonso-Pinar et al., 2025).

Coupled simulations are designed to run efficiently on a supercomputing cluster but generates hundreds of gygabyte data file , so we added in ForeFire Pakage some Python helper scripts available in the ForeFire repository convert these simulation outputs to VTK/VTU files, allowing three-dimensional rendering of fire-atmosphere variables in Open-Source ParaView as shown in Figure 2.

Coupled simulations are designed to run efficiently on supercomputing clusters, but they generate hundreds of gigabytes of output data. Python helper scripts are available in the ForeFire repository to facilitate post-processing by converting these outputs into VTK files for open-source visualization tool ParaView, as illustrated in Figure 2.

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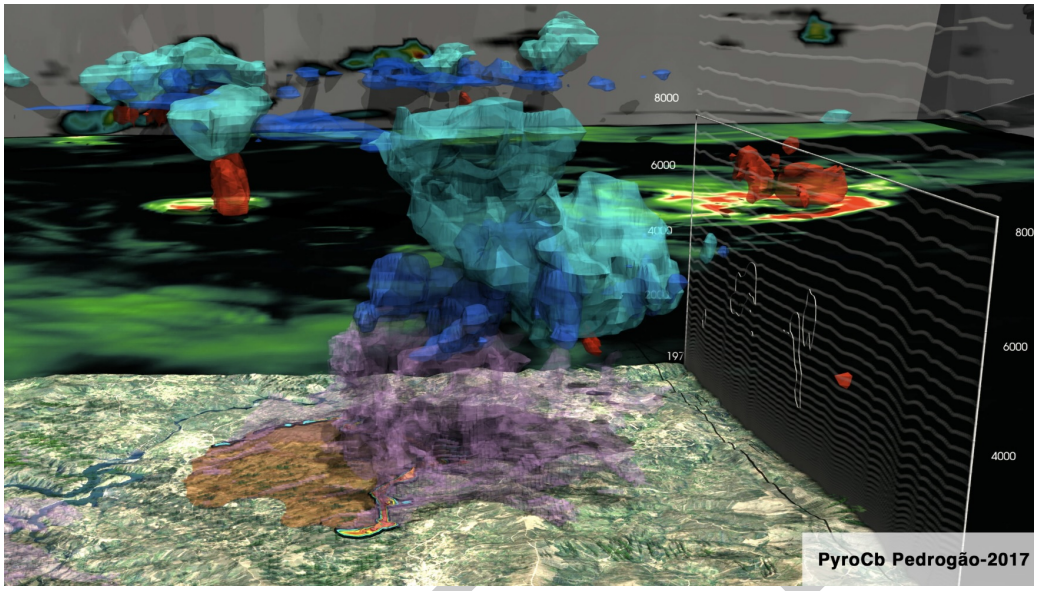

Figure 2: Coupled simulation of the Pedrogao Grande wildfire (Couto et al., 2024), rendered in Paraview. On the ground, the burned area is in orange, while among atmospheric variables, downbursts are highlighted in red and pyro-cumulonimbus clouds in blue.

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